

Comparison of lab-scale experiments for the determination of non-reactive tracer diffusion coefficients in Boom Clay

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ABSTRACT

The transport parameters of radionuclides, the diffusion coefficient and the retardation factor, in clay-rich host rocks investigated for deep geological disposal, can be determined using various types of laboratory experiments. The present study aimed at comparing three types of laboratory experiments commonly used in the context of radioactive waste for the determination of the transport properties of two non-reactive tracers, HTO and ^{125}I , in Boom Clay. Two Boom Clay samples with orientation parallel and perpendicular to the bedding were selected to assess the role of sample heterogeneity and anisotropy. Each sample/orientation was subjected to through-diffusion experiments, pulse injection experiments with pressure differences over the clay core ranging from 5 to 19 bars and a back-to-back experiment.

In the case of HTO, the three types of experiments led to a consistent set of transport parameters for each Boom Clay sample. Over the three types of experiment, the average apparent diffusion coefficients and accessible porosity vary at maximum a factor 1.4 and 1.3, respectively for a same sample type/orientation. The porosity accessible to HTO was similar to the porosity calculated from the water content. In the case of ^{125}I , the three types of experiments also showed consistency for the determination of the apparent diffusion coefficients with values varying within a factor 1.7 for the same sample type/orientation. However, variation in accessible porosity between the experiments revealed a physical modification of the Boom Clay sample occurring during the pulse injection experiments and affecting ^{125}I transport.

Transport anisotropy was consistently observed in all the types of experiment and for both HTO and ^{125}I with a higher anisotropy observed in the samples richer in clay. Slightly higher total porosity and lower ^{125}I accessible porosity were noticed in the clay-rich samples in comparison to silt-rich samples.

1. Introduction

Several countries have opted for deep geological disposal for the long-lived and high activity nuclear wastes such as Belgium (www.niras.be, www.ondraf.be), France (www.andra.fr), Swiss (www.nagra.ch), Sweden (www.skb.com) or Finland (www.stuk.fi/en/nuclear-waste-management). For the development of safety cases, experimental values of transport parameters in host rocks are essential for both unretarded and retarded radionuclides. An accurate determination of these parameters requires reliable experimental approaches representative of the real system. As such, large scale experiments performed *in-situ* can be regarded as the best approach (Put et al., 1993; Van Loon et al., 2012; Weetjens et al., 2014). Their localization directly ‘on-site’ limits external disturbances while their scale ensures a good

representativeness of the possible heterogeneity of the rock formation/layer. However, setting-up large scale *in-situ* experiments is technically challenging. Working *in-situ* requires an access to an underground laboratory or to a deep borehole. It also means injecting radioactive tracers directly in the natural environment, which is strictly regulated in most countries. To minimize the disturbances of the system, on-line measurements should be favoured over out-line measurements and installed directly in the borehole or underground laboratory. Working over large scales requires excavation and experimental set-ups of several meters. Though some underground facilities have been used for *in-situ* tracer tests (Aertsens et al., 2022; Delay et al., 2007; Gimmi et al., 2014; Trinchero et al., 2020), the costs to set up and perform the experiments have limited their number. For most host rocks considered, including Boom Clay studied in Belgium (Bruggeman and Maes, 2017),

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most of the experiments on the transport of radionuclides were performed in the laboratory on rock samples of a couple of centimetres. Lab-scale experiments have indeed the advantage to be easy to set-up and to be fast in comparison to large scale *in-situ* experiments (Van Loon et al., 2012) allowing to investigate a variety of radionuclides and conditions: effect of mineralogical composition (Jacops et al., 2017; Savoye et al., 2014), ionic strength (Aertsens et al., 2009; Glaus et al., 2015; Savoye et al., 2015; Wittebroodt et al., 2012), temperature (Joseph et al., 2013), inorganic or organic ligands (Joseph et al., 2013; Maes et al., 2011; Trancom-Clay, 2000) ...

In clay-rich host rocks and in undisturbed conditions, transport of radionuclides is dominated by diffusion and the important radionuclide transport parameters are the diffusion coefficient and the rock capacity factor, the product of the accessible porosity and the retardation factor. One can assume that pure diffusion lab-scale experiments are the most representative of *in-situ* transport and the most suited for the determination of the radionuclide transport parameters (Van Loon et al., 2012). Yet because of the time this kind of experiments can take, especially with retarded radionuclides, some research institutes have also used other type of experiments with either a pressure or electrical gradient. Hence a variety of lab-scale transport experiments were used for the determination of the diffusion coefficients and the rock capacity factors of radionuclides in clays: through-diffusion (TD) (Altmann et al., 2015; Descostes et al., 2008; García-Gutiérrez et al., 2004; Jacops et al., 2017; Van Loon et al., 2004, Van Loon et al., 2005a), in-diffusion (Altmann et al., 2015; Van Loon and Eikenberg, 2005), back-to-back/closed cell/instantaneous planar source/thin source (BtB) (Aertsens et al., 2008; Bourg and Tournassat, 2015; García-Gutiérrez et al., 2011), percolation or pulse injections (PI) (Aertsens et al., 2004, 2008) and electromigration experiments (Maes et al., 1999; Tanaka et al., 2011). Each type of experiment could have its artifacts possibly affecting the determination of the transport parameters: presence of filter plates, pressure gradient, electrical gradient, dead volumes, etc. The models used to extract the transport parameters can to a certain extent take these artifacts into account, but their complexity could also be a source of inconsistencies. Some studies have focused on these artifacts and evaluated their possible effects (Aertsens et al., 2020; Glaus et al., 2007; Suzuki et al., 2003; Van Laer et al., 2023). Benchmark studies between institutes were also performed to evaluate the robustness of a particular experimental approach (Van Laer et al., 2023). However, though these works gave more insight on each experimental approach individually, the consistency of different approaches for the determination of transport parameters was rarely assessed. Aertsens et al. (2008) have looked into the consistency of percolation and in-diffusion experiments for the transport of ^{32}Si in Boom Clay. They noticed some variation between the ^{32}Si transport parameters obtained with the two experimental approaches especially for the rock capacity factor. However, the authors worked with different Boom Clay samples and sample heterogeneity could have played a role. A good consistency study requires avoiding any sample heterogeneity effect, which could complicate the interpretations. A systematic study in which various experimental approaches are tested on the same set of samples is therefore important to ensure the validity of the determined transport parameters. This is particularly true in the case of Boom Clay, for which several experimental approaches were used over the years. A large part of the experimental transport parameters of the non-retarded tracers HTO and ^{125}I was obtained from pulse injection experiments while through-diffusion experiments have been used in the recent years as the reference method for unretarded or slightly retarded tracers. Due to the imposed pressure gradient, pulse injection experiments are potentially less representative and more difficult to interpret than pure diffusion experiments. It was indeed shown by Aertsens et al. (2020) that the model previously used for the determination of the iodine rock capacity factor from pulse injection experiments was erroneous.

Therefore, the present work has for objective to investigate the consistency of the transport parameters determined by pure diffusion

type transport experiments and pulse injection experiments on a single sample set. This study focuses on the non-reactive tracers HTO and ^{125}I , which were injected simultaneously in all experiments. Two types of pure diffusion experiments were tested: through-diffusion experiments and what will be referred to as back-to-back experiments. Through-diffusion experiments consist of bringing the rock sample in contact with two solution reservoirs, one containing the tracer and one (initially) tracer-free (Aertsens et al., 2008; Van Loon et al., 2012) and was largely used in the last decades for the determination of radionuclide transport parameters in clay-rich rock. Back-to-back experiments, referred to as closed cell (Bourg and Tournassat, 2015) or instantaneous planar source method (García-Gutiérrez et al., 2011), was less used because they allow the determination of only the apparent diffusion coefficient and not of the rock capacity factor. Yet they have the advantage to avoid any effect of the filter plate. In these experiments, the tracer is indeed injected in the centre of the core and the experiment is stopped before its breakthrough out of the core. Pulse injection experiments as referred to by Aertsens et al. (2008), are a variation of the traditional percolation experiments (Durge et al., 2014, 2018; Latrille et al., 2007). Transport in these experiments is a mix of advection and diffusion and as such, they do not allow a direct determination of the diffusion coefficient but of the dispersion coefficient. In a 1D system, diffusion and dispersion coefficients are linked by:

$$D_{app}^i = \alpha_L V_{app} + D_{app} \quad (1)$$

with V_{app} the tracer apparent velocity (m/s), D_{app}^i the tracer apparent dispersion coefficient (m^2/s), D_{app} the tracer apparent diffusion coefficient (m^2/s) and α_L the dispersion length (m). Although at low velocity and/or for systems with low dispersion, dispersion coefficients approximate diffusion coefficients, an accurate determination of the diffusion coefficient requires an estimation of the dispersion length. To that intend and to allow comparison with pure diffusion experiments, a series of pulse injection experiments at different hydraulic gradients was performed here. Boom Clay is anisotropic and locally heterogeneous. The orientation of the samples to the bedding and their mineral assemblage were shown to lead to different transport parameters for HTO (Aertsens M. et al., 2013; Jacops et al., 2017). The dispersion length is sample-dependent and could vary with the orientation of the samples and their mineral assemblage. Considering these possible variations, the consistency between the three experimental approaches was investigated on clay-rich as well as on silt-rich/homogeneous samples and between samples with orientation parallel and perpendicular to the bedding. The full set of performed experiments, which is indicated in Figure ES1, spread over a period of 3.5 years. The experimental work was described in Durce and Van Gompel, (2022a) and the present paper is partly based on that report.

2. Materials and methods

2.1. HTO and ^{125}I solutions and radiometric analyses

A solution of ^{125}I in 1×10^{-5} M NaOH was purchased from PerkinElmer with a volumetric activity of 3.7×10^9 Bq/mL (at the date of purchase). The source was a mix I^-/IO_3^- . Before use of the source, IO_3^- was reduced to I^- in 10^{-3} M of Na_2SO_3 first acidified to pH 2.

The activity of ^{125}I in solution was determined by gamma counting on a Packard COBRA (type 5003) gamma counter with a counting efficiency estimated at 81 %.

The activity of HTO was determined by liquid scintillation on a Packard TRI-CARB 2100 TR counter in the region 0–12 keV. HTO and ^{125}I being injected simultaneously in all experiments, the presence of ^{125}I interfered with HTO counting in the region 0–12 keV and the contribution of ^{125}I to the total counts had to be subtracted. It was estimated that the efficiency of ^{125}I counting in the region 0–12 keV relative to the activity determined by gamma spectrometry was 61.3%.

The measured counts per volume (CPM/mL) of HTO was calculated according to Equation (2) and the counting efficiency was estimated to 57 %

$$A^{LS}(\text{HTO}) = \frac{(CPM_{\text{sample}}^{LS} - CPM_{\text{bkgd}}^{LS})}{V^{LS}} - A^{GC}(\text{I125}) * 0.613 \quad (2)$$

with CPM_{Sample}^{LS} and CPM_{bkgd}^{LS} the number of counts (CPM) measured in liquid scintillation (LS) in the sample and in the background solution respectively, V^{LS} the volume of sample (mL) used for the measurement in liquid scintillation and $A^{GC}(\text{I125})$ the number of count per volume (CPM/mL) of ^{125}I measured by gamma counting (GC).

All the activities of HTO and ^{125}I measured in solution are here reported as number of counts per minute and per volume, i.e., CPM/mL.

2.2. Boom Clay pore water

Boom Clay pore water was collected from the four filters of the horizontally oriented piezometer TD-116E, installed at the level of the HADES laboratory (−196.6 m Tweede Algemene Waterpassing, Mol, Belgium). It was filtered at 0.45 μm (PVDF hydrophilic, VWR) before use. The composition of the water was not measured in this work but the composition reported by Durce et al., (2022b) is given in the Electronic Supplement (Table ES1).

2.3. Clay samples

Two Boom Clay samples were cored from two different depths (Members) of the Boom Clay. One sample was cored from the clay-rich Putte Member region, and one from the Boeretang Member, richer in silt and previously referred to as the ‘transition zone’ (Aertsens et al., 2004). The samples were taken from the ON-MOL-1 borehole, which was drilled in 1997 in Mol in the northeast of Belgium (Aertsens et al., 2004). Prior to sampling, the core material was stored under vacuum in aluminium foil bags at a temperature of 4 °C. From each sample, four clay plugs were cut with dimensions $\sim 38 \times 32$ mm ($\phi_{\text{plug}} \times L_{\text{plug}}$), two with an orientation parallel to the bedding and two with an orientation perpendicular to the bedding. The samples were characterized by XRD, laser diffraction, Total organic content (TOC) analyser, N_2 -adsorption measurements and Mercury Intrusion Porosimetry. The results of all the measurements can be found in Durce et al., (2022b). Other general properties of Boom Clay can also be found elsewhere (De Craen, 2005; Frederickx, 2020; Honty and De Craen, 2011; Wang et al., 2023). We report here, in the Electronic Supplement (Table ES3), only the average mineralogical composition of the bulk samples and of the clay fraction as determined by quantitative XRD by Q-Minerals (Leuven, Belgium) on subsamples taken in the vicinity of the four clay plugs. The determined composition is in agreement with the mineralogical differences of the two Boom Clay regions from which the samples are extracted. Hence, the sample taken from the Putte Member shows a higher clay-mineral content, an average of 67 %, in comparison to the sample taken from the Boeretang Member, which contains on average 43 % of clay minerals.

The hydraulic conductivity of the samples was determined before the start of the experimental series. During coring and subsequent storage and cutting, the clay plugs were expected to (partially) dehydrate. The plugs were resaturated by percolating Boom Clay pore water under a pressure of 5 bars. When constant flow conditions were established the hydraulic conductivity K_H (m/s), was calculated according to Equation (3):

$$K_H \approx \frac{\bar{Q} \times L}{10\Delta P \times A} \quad (3)$$

with $\bar{Q}$ the average flow rate (m^3/s), L the length of the clay plug (m), ΔP the pressure difference (bar) and A the area of the clay plug (m^2).

The hydraulic conductivities of the eight clay plugs are reported in Table 1. For the plugs taken from the Putte Member region, higher hydraulic conductivities were determined for the plugs with an orientation parallel to the bedding than for the ones oriented perpendicularly. This is in agreement with the Boom Clay anisotropy (Aertsens et al., 2004). The sample Boer-Pa1 displayed a significantly higher hydraulic conductivity than the others. This could be the result of small fractures generating preferential path or higher connectivity between sand and silt regions. The sample showed irreversible damages with the presence of a crossing-fracture after the through-diffusion experiments, making further experiments on this sample useless.

2.4. Through-diffusion experiments

The principle and set-ups of through-diffusion experiments have been largely detailed in the literature (Aertsens et al., 2008; Descostes et al., 2008; García-Gutiérrez et al., 2004; Van Loon et al., 2003) and only the specificities of the used experimental set-up will be given here. In the through-diffusion experiments performed in this work, the inlet tracer concentration was let to vary as a function of time while the outlet tracer concentration was maintained closed to zero by replacing the outlet solution with tracer-free solution after each measurement. The tracer activity and volume of the inlet solution were chosen large enough to maintain a concentration gradient in the sample and to limit the effect of sampling. The set-up is shown in Fig. 1.

Two through-diffusion experiments were performed on each plug: one before the series of pulse injection experiments and one after (Figure ES1) to evaluate whether the pulse injection experiments affected the clay samples. The two experimental sets are further referred to as TD1 and TD2, respectively.

The eight clay plugs were loaded into stainless steel diffusion cells, between two stainless steel filter plates. Each cell was connected to an inlet reservoir filled with 200 mL Boom Clay pore water spiked with ~ 100 – 200 Bq/mL HTO in TD1 and TD2 and ~ 200 – 300 Bq/mL ^{125}I in TD1 and ~ 50 Bq/mL ^{125}I in TD2, while the outlet reservoir contained 20 mL of pure Boom Clay water. Outlet solutions were sampled and replaced regularly and aliquots of the inlet were sampled at the same frequency. The frequency of sampling varied from experiments to experiments and along the time scale of the experiments as can be seen in Figure ES2 to ES9. The diffusion of the tracers was followed by measuring the outlet and inlet samples by liquid scintillation counting and gamma counting as described in the previous section. The experiments were continued until the outlet flux of both tracers was approximately constant (see Figures ES2 to ES9). To ‘clean-up’ the plugs before the pulse injection series, after the TD1 experiments, the cells were connected to a percolation set-up (Fig. 1 b) and the tracers were percolated out at a pressure gradient of 5 bars. During this phase, outlet samples were taken only sporadically to control the progress of the out-percolation and these results are not presented here. After the series of

Table 1

Information on the Boom Clay Plugs. L_{plug} = plug length, ϕ_{plug} = plug diameter, K_H = hydraulic conductivity (Equation (3)).

Reference	Origin	Orientation to the bedding	L_{plug} (mm)	ϕ_{plug} (mm)	K_H (10^{-12} m/s)
Putte-Pa1	Putte Member	Parallel	31.8	38	2.6 ± 0.5
Putte -Pa2		Parallel	31.8	38	3.6 ± 0.1
Putte -Pe1		Perpendicular	31.6	38	1.7 ± 0.1
Putte -Pe2		Perpendicular	31.6	38	1.5 ± 0.1
Boer-Pa1	Boeretang Member	Parallel	31.9	38	6.8 ± 0.2
Boer -Pa2		Parallel	32.0	38	2.9 ± 0.2
Boer -Pe1		Perpendicular	31.9	38	2.8 ± 0.1
Boer -Pe2		Perpendicular	31.9	38	3.9 ± 0.1

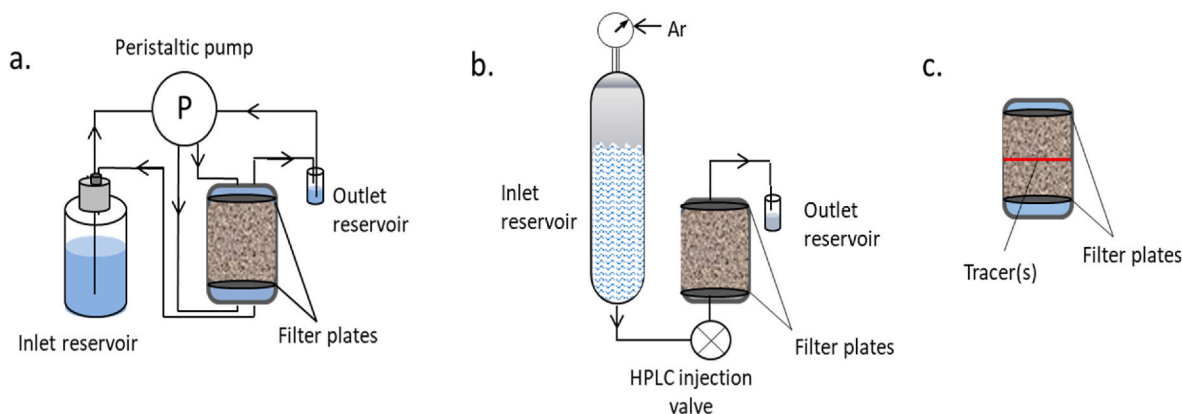


Fig. 1. Experimental set-ups. a. Through-diffusion set-up b. Pulse injection set-up c. Back-to-back set-up. Filter plates = GKN Filter Technology, SIKA-R 5 AX, average pore diameter 11 μm , thickness 2 mm.

pulse injection experiments, for which the samples were transferred into percolation cells, the samples were transferred back to the diffusion cells and the TD2 experiments performed.

2.5. Pulse injection experiments

The set-up used here and reported in Fig. 1 was similar to the set-up described by Aertsens et al. (2008). HTO and ^{125}I were injected simultaneously using a 20 μL HPLC injection valve positioned before the cell containing the sample. The inlet solution, which percolated through the Boom Clay plug, was Boom Clay pore water pressurized with Ar gas.

The design of the percolation cells differed slightly from the design of the diffusion cells and after the through-diffusion experiments, the Boom Clay plugs were transferred from diffusion cells to percolation cells. The transfer was done in air but contact time of the Boom Clay plugs with the atmosphere was limited to a couple of minutes preventing dehydration and clay oxidation.

A series of pulse injection experiments was performed sequentially at different pressure differences: 5, 7, 10, 12, 15 and 19 bar. The experiments were completed when the activities of ^{125}I and HTO in the outlet were close to zero. Then, the hydraulic gradient was changed and after stabilization of the outlet flow rate, the tracers were reinjected using the HPLC injection valve. The injected activities ranged from ~ 200 to ~ 1000 Bq for ^{125}I and from ~ 400 to ~ 700 Bq for HTO (Table ES9). The large range of activities used for ^{125}I is due to the significant decay (half-life ~ 60 days) of the spike solution occurring over the time scale of the study. The breakthrough of the tracers was followed by regularly collecting and analysing the outlet solutions by liquid scintillation counting and gamma counting.

2.6. Back-to-back experiments

As a last step, the transport of HTO and ^{125}I was followed in back-to-back experiments. Back-to-back experiments were performed with the experimental set-up reported in Fig. 1 c.

Each clay plug was retrieved from the cell and cut into two sub-plugs of ~ 16 mm. The exact size of all the sub-plugs is reported in the Electronic Supplement (Table ES4). A first sub-plug was reintroduced in a diffusion cell and 200 μL of a solution containing 7.8×10^3 Bq of HTO and $4.7\text{--}5.1 \times 10^3$ Bq of ^{125}I was uniformly deposited on the plug surface. The second sub-plug was loaded in the cell and pushed to contact with the tracer solution and the first sub-plug before closing the cell. The cells were then left on a table standing vertical for 8–20 h depending on the experiment (Table ES4). The experimental time was estimated from the results of the through-diffusion experiments and chosen to ensure enough diffusion of the tracers without allowing their breakthrough. At the end of the experimental time, the clay plugs were extruded from the

cells and cut into slices of 0.5–2 mm. Each slice was divided into three pieces. One piece was analysed for ^{125}I activity by gamma counting, one piece was oven dried at 60 $^{\circ}\text{C}$ to calculate the water content and one piece was further leached out in ultrapure water and analysed for HTO activity by liquid scintillation counting.

2.7. Data presentation and modelling

All the data are reported in CPM/mL or CPM/cm but can be converted to Bq/mL or Bq/cm assuming a counting efficiency of 81 % for ^{125}I and 57 % for HTO. The radioactive decay was taken into account and all activities were recalculated to zero time. Experimental uncertainties were calculated by propagating the experimental errors (confidence limit of 95 %). The errors on the values obtained by fitting are given at 1 σ (68 %).

The results of the modelling at each discrete experimental point are presented in the Electronic Supplement together with the experimental data. It should be noted that to ease the reading of the figures the modelled points are not plotted but only the connecting lines are.

2.7.1. Through-diffusion experiments

Two transport parameters can be obtained from through-diffusion experiments: the apparent diffusion coefficient, D_{app} and the rock capacity factor ηR , being the product of the accessible porosity η and the retardation factor R . These two parameters were determined by solving the diffusion equation (Equation (4)) with the appropriate boundary and initial conditions (variable inlet concentration, zero outlet concentration, see Electronic Supplement) by using the software COMSOL Multiphysics 3.5a, Earth science module using the MATLAB R2012a interface:

$$\frac{\partial C}{\partial t} = D_{app} \frac{\partial^2 C}{\partial x^2} \quad (4)$$

with C the tracer concentration/activity (CPM/mL) and D_{app} the tracer apparent diffusion coefficient (m^2/s).

The rock capacity factor (ηR) and the apparent diffusion coefficient (D_{app}) were optimized by fitting simultaneously inlet and outlet activity evolution by minimizing the sum of squared residuals using the non-linear least square method in the MATLAB Optimization Toolbox. The residuals were expressed relative to the measured values according to Equation (5).

$$\chi_r^2 = \sum_{i=1}^N \left(\frac{f(x_i, P) - y_i}{y_i} \right)^2 \quad (5)$$

with x a variable (time in the present case), P the parameters (ηR and D_{app}), $f(x_i, P)$ the model prediction (for both inlet and outlet data), and

the index i varying between one and the number of points N . The minimization of relative residuals and not absolute residuals was necessary due to the difference in inlet and outlet tracer activities. However, one should keep in mind that, with this approach, outlet data have more weight in the fitting procedure compared to inlet data, which are in consequence less well depicted.

The filters were integrated in the model by assuming that they display the same transport properties as the clay (Aertsens, 2011).

2.7.2. Pulse injection experiments

Due to the imposed hydraulic gradient, a flow is present in pulse injection experiments and transport is a combination of diffusion and advection. The experimental breakthrough curves are modelled by solving the advection-dispersion equation (Equation (6)):

$$\frac{\partial C}{\partial t} = -V_{app} \frac{\partial C}{\partial x} + D_{app}^i \frac{\partial^2 C}{\partial x^2} \quad (6)$$

with C the tracer concentration/activity (CPM/mL), V_{app} the tracer apparent velocity (m/s) and D_{app}^i the tracer apparent dispersion coefficient (m²/s).

The software COMSOL Multiphysics 3.5a, Earth science module with the MATLAB R2012a interface was used and the apparent velocity (V_{app}) and the apparent dispersion coefficient (D_{app}^i) were optimized by minimizing the absolute residuals, $\chi_a^2 = \sum_{i=1}^N (f(x_i, P) - y_i)^2$. The rock capacity factor was calculated according to Equation (7):

$$V_{Darcy} = \eta R V_{app} \quad (7)$$

with V_{Darcy} (m/s) the Darcy velocity.

It was demonstrated by Aertsens et al. (2020) that in the case of anionic tracers, Equation (7) is not correct. An accurate determination of the iodine accessible porosity requires some knowledge on the fraction of V_{Darcy} flowing through the iodine accessible porosity. This information cannot be accessed with the data here obtained and for the sake of comparison and discussion, it was chosen to keep using Equation (7) to determine the iodine rock capacity factor but keeping in mind its inaccuracy.

More details on the model can be found in Aertsens et al., (2003a,b). The model was extended with filters following the same approach as for the through-diffusion model. Initially, all tracer is assumed at the interface between filter at the inlet reservoir side and the tube leading to the inlet filter.

2.7.3. Back-to-back experiments

In case of the back-to-back experiments, the activity profiles of ¹²⁵I and HTO along the plugs were presented in CPM/cm as a function of the distance from the spiking point. They were determined according to Equations ES1 to ES5 reported in the Electronic Supplement.

Contrary to through-diffusion experiments, back-to-back experiments allow only the determination of the tracer apparent diffusion coefficient D_{app} . This was done by solving Equation (8) with COMSOL Multiphysics 3.5a

$$C(x, t) = \frac{Q_0}{2 \eta R S \sqrt{\pi D_{app} t}} \exp\left(-\frac{x^2}{4 D_{app} t}\right) \quad (8)$$

with Q_0 the initially injected tracer quantity (CPM), $S = (\pi/4) \varnothing^2$ the clay core cross section surface and $\varnothing$ the diameter of the core (cm) as reported in Table 1. Experimentally, the amount of tracer per slice and the slice thickness and mass are measured, leading to the concentration (activity) per cm clay (see Electronic Supplement, Figures ES10 to ES13), which is proportional to $C(x, t)$.

The average total porosity of the plugs was determined from the water content of each slice, WC (Equation ES1) according to Equation (9).

$$\eta_{tot} = \frac{\sum_{i=1}^N \frac{4}{l_{s,i}} \frac{WC_i}{\varnothing^2} \frac{m_{s,wet,i}}{\rho_w}}{N} \quad (9)$$

with WC_w , the water content (–) of the slice, $m_{s,wet}$ the weight (g) of the wet slice, l_s (cm) the thickness of the slice, ρ_w the density of water (1 g/cm³) and N the number of slices.

3. Results and discussion

HTO and ¹²⁵I are assumed non-reactive tracers and the rock capacity factor ηR is taken equal to the accessible porosity, η (so the retardation factor $R = 1$).

The transport parameters of ¹²⁵I and HTO obtained for each sample, in each type of experiments are discussed and compared in the following sections. They are all summarized in Table 4

3.1. Pure diffusion experiments: through-diffusion and back-to-back experiments

3.1.1. Through-diffusion: TD1 experiments

The experimental and modelled inlet and outlet activity and flux over time for HTO and ¹²⁵I obtained for the eight Boom Clay plugs are reported in the Electronic Supplement (Figures ES2 to ES9). The transport parameters obtained from the best fit of the experimental data are summarized in Table 2.

As expected, the apparent diffusion coefficient and the accessible porosity are higher for HTO than for ¹²⁵I for all samples. This is

Table 2

Fitted transport parameters of ¹²⁵I and HTO in through-diffusion experiments samples before (TD1) and after (TD2) the series of pulse injection experiments. After the pulse experiments, sample Boer-Pa1 was damaged and no further experiments were performed on it. The uncertainty values correspond to the numerical error.

Sample reference	¹²⁵ I		HTO	
	D_{app} (10 ^{–10} m ² /s)	η	D_{app} (10 ^{–10} m ² /s)	η
Putte-Pa1/TD1	3.49 ± 0.05	0.09 ± 0.01	5.54 ± 0.10	0.32 ± 0.01
Putte-Pa2/TD1	3.37 ± 0.05	0.09 ± 0.01	5.52 ± 0.11	0.34 ± 0.01
Putte-Pe1/TD1	1.49 ± 0.03	0.06 ± 0.01	2.69 ± 0.03	0.32 ± 0.01
Putte-Pe2/TD1	1.38 ± 0.03	0.10 ± 0.01	2.38 ± 0.01	0.38 ± 0.01
Boer-Pa1/TD1	2.62 ± 0.25	0.17 ± 0.02	6.14 ± 0.10	0.33 ± 0.02
Boer-Pa2/TD1	2.87 ± 0.08	0.11 ± 0.01	3.37 ± 0.44	0.63 ± 0.09
Boer-Pe1/TD1	1.97 ± 0.03	0.10 ± 0.01	3.15 ± 0.07	0.39 ± 0.02
Boer-Pe2/TD1	1.83 ± 0.03	0.16 ± 0.01	3.94 ± 0.10	0.33 ± 0.02
Putte-Pa1/TD2	2.65 ± 0.10	0.16 ± 0.01	4.73 ± 0.25	0.33 ± 0.03
Putte-Pa2/TD2	2.71 ± 0.08	0.18 ± 0.01	5.51 ± 0.24	0.29 ± 0.02
Putte-Pe1/TD2	0.98 ± 0.08	0.32 ± 0.04	2.61 ± 0.10	0.29 ± 0.02
Putte-Pe2/TD2	0.84 ± 0.04	0.24 ± 0.03	2.47 ± 0.08	0.34 ± 0.02
Boer-Pa1/TD2	Experiment not performed			
Boer-Pa2/TD2	2.46 ± 0.07	0.21 ± 0.01	5.05 ± 0.08	0.27 ± 0.01
Boer-Pe1/TD2	1.37 ± 0.09	0.20 ± 0.03	3.24 ± 0.05	0.29 ± 0.01
Boer-Pe2/TD2	1.67 ± 0.03	0.20 ± 0.01	3.62 ± 0.07	0.27 ± 0.01

characteristic of the anion exclusion phenomenon occurring in clay materials due to the permanent negative charge bore by the clay sheets (Aertsens et al., 2020; Van Loon et al., 2007).

Only few through-diffusion experiments were reported on Boom Clay samples but the apparent diffusion coefficients determined here for HTO for the TD1 set and for a perpendicular orientation are lower than those reported by Jacops et al. (2017) ($D_{app} \approx 3.2\text{--}4.3 \cdot 10^{-10} \text{ m}^2/\text{s}$ (Putte) and $\approx 3.8\text{--}4.9 \cdot 10^{-10} \text{ m}^2/\text{s}$ (Boeretang)). Samples in that study displayed higher hydraulic conductivities than the samples used here. The experimental set-up, which was intended for measuring gas diffusion, was also slightly different than here. This could explain the observed differences between the two studies.

Due to the simultaneous injection of ^{125}I and HTO and the resulting interference of ^{125}I with HTO in liquid scintillation counting, the experimental errors on HTO data points are high. This is especially true for high injected ^{125}I activities as it was the case for the 'Boer' experiments (Figure ES2). In comparison with the 'Putte' experiments, the 'Boer' experiments also show data that are more scattered and less homogeneity between the two plugs with the same flow orientation. This is consistent with the hydraulic conductivity of the Boer plugs (Table 1) which is less reproducible especially for the two plugs Boer-Pa1 and Boer-Pa2. The experimental inlet concentrations for the two plugs increase after 60 days (Figure ES2), which is not expected. The reason for this increase remains unclear but could be related to sampling or counting though no clear evidence was found. Evaporation was ruled out as the same increase was not visible in the ^{125}I concentration. The model allows a good description of most of the experimental data. For Boer-Pa2, HTO (Figures ES2), the low activities of the first breakthrough points lead for the relative optimization to a very low D_{app} and an unrealistic $\eta = 1.11$. These first breakthrough points are less determining for the optimal values resulting from an absolute fit, resulting in a more realistic but still anomalous D_{app} and η value in comparison to Boer-Pa1 and the other plugs (Table 2). These individual values for D_{app} and η are considered as erroneous and omitted from the rest of the paper and of the discussion.

Excluding the Boer-Pa samples, an overall good reproducibility is observed for the apparent diffusion coefficients of both tracers with a maximum difference of 25 % between the two experimental repeats. The accessible porosity appears as a more sensitive parameter, less reproducible, especially in the case of ^{125}I , where up to 63 % difference is obtained between two repeats.

The two Boer-Pa plugs displayed an enhanced hydraulic conductivity after the through-diffusion experiments with values 9 and 6 times higher for Boer-Pa1 and Boer-Pa2 than initially measured, strongly pointing to sample damage. The origin of the damage was not elucidated but since the tracer profiles and the transport parameters obtained during the through-diffusion experiments are mostly in line with the other experiments, it is assumed that the samples got damaged more towards the end of the through-diffusion experiments. The plugs were allowed to self-seal by imposing a low hydraulic gradient, i.e., 1 bar, during the out-percolation phase (Figure ES1). Boer-Pa2 recovered its initial hydraulic conductivity but though the Boer-Pa1 sample partially sealed, it never recovered its initial hydraulic conductivity and it was decided to not perform further experiments on this sample.

3.1.2. Through-diffusion: TD2 experiments

A second series of through-diffusion experiments, TD2, was performed after the series of pulse injection experiments to assess the possible effect of the hydraulic gradients on the Boom Clay plugs. The results of this second series are compared to the first series of experiments in Table 2. It should be noted that due to decay, the activities of ^{125}I injected in this second series were low leading to low to very low outlet activities (Figures ES8 and ES9), making the modelling exercise more uncertain, the initial breakthrough being more difficult to detect. However the good reproducibility between repeats was still maintained with a maximum variation between repeats of 22 % and 32 % for D_{app}

and η , respectively. For both tracers, the transport parameters show some variation with respect to the ones determined before the pulse injection experiments (TD1). In the case of HTO, similar apparent diffusion coefficients were determined in the second through-diffusion experimental series but slightly lower accessible porosities. On the other hand, in the case of ^{125}I , higher accessible porosities but lower apparent diffusion coefficients were determined after the pulse injections. The differences between the two experimental series are systematic in all the samples. This suggests that the series of pulse injection experiments had an effect, even if limited, on the Boom Clay plugs.

3.1.3. Back-to-back experiments

The back-to-back experiments, which were performed after the second series of through-diffusion experiments, display a poorer reproducibility between repeats than the latter. The apparent diffusion coefficients determined on the same sample type differ from one another from 2 to 65 % for HTO and from 27 to 85 % for ^{125}I (Table 3). The poor reproducibility reflects the fragility of the experimental and modelling results. Experimental profiles (Electronic Supplement, Figure ES10 to Figure ES13) reveal scattered data especially close to the point of spiking. Around that point, the activity of both tracers decreases sharply and the model does not capture the peak. For some experiments, the activity peak is also slightly shifted from the zero coordinate. The thickness of the clay slices did not provide a good resolution of the peak and the re-sealing of the sub-plugs prevented the localization of the point of spiking, which had to be calculated leading to more uncertainty on the position of the peak. The discrepancy between the model and the experimental results around the spiking point brings more uncertainty on the determined values of D_{app} , though an accurate evaluation of this uncertainty remains difficult. This is especially true for iodide for which the model shows the strongest deviation with the experimental data.

On the other hand, the total porosity values calculated from the measured water content are well reproducible (Table 3). The total porosity is, as expected isotropic, but appears lower for the plugs originating from the Boeretang Member than for the plugs taken from the Putte Member. Jacops et al. (2017) did not explicitly point out a water content difference between clayey and silty Boom Clay samples though the authors reported a slightly lower total porosity for samples originating from the Boeretang Member than for the samples originating from the Putte Member. Clay minerals displaying higher intraparticle porosity than silts, this difference is in agreement with the mineralogy of the plugs and with the results of Hemes et al. (2013) who also reported a higher total porosity for a higher clay content of the Boom Clay samples. Table 4

3.2. Pulse injection experiments and effect of hydraulic gradient

A large number of pulse injection experiments (a total of 40) was performed in this work and only a few experimental and modelled breakthrough curves are reported in the Electronic Supplement

Table 3

Fitted apparent diffusion coefficients of ^{125}I and HTO and calculated total porosity from water content measurements at the end of back-to-back experiments. Sample Boer-Pa1 was damaged, so no experiments were performed on it. The uncertainty values for D_{app} correspond to the numerical error (1 σ) and for η_{tot} correspond to the standard deviation.

Sample Reference	$D_{app}^{125}\text{I}$ ($10^{-10} \text{ m}^2/\text{s}$)	D_{app} HTO ($10^{-10} \text{ m}^2/\text{s}$)	η_{tot}
Putte -Pa1/BtB	2.67 ± 0.14	6.91 ± 0.37	0.38 ± 0.02
Putte -Pa2/BtB	1.50 ± 0.25	7.01 ± 0.50	0.36 ± 0.01
Putte -Pe1/BtB	0.91 ± 0.23	2.61 ± 0.17	0.35 ± 0.07
Putte -Pe2/BtB	1.68 ± 0.56	4.30 ± 0.96	0.33 ± 0.03
Boer -Pa1/BtB	Experiment not performed		
Boer -Pa2/BtB	3.06 ± 0.59	5.33 ± 0.71	0.28 ± 0.01
Boer -Pe1/BtB	1.10 ± 0.06	2.91 ± 0.13	0.30 ± 0.02
Boer -Pe2/BtB	1.40 ± 0.10	3.73 ± 0.23	0.30 ± 0.01

Table 4

Summary of the determined transport parameters for ^{125}I and HTO. Values in *italic* are assumed erroneous and neglected. PI = pulse injection series, TD1 = through-diffusion experiments performed before the pulse injections, TD2 = through-diffusion experiments performed after the pulse injections. D_{app} = apparent diffusion coefficient, η = accessible porosity, η_{tot} = total porosity calculated from water content. * = average of the two plugs with the spread or standard deviation. ** = values from one plug given with the numerical or measurement error. *** = extrapolated to zero Darcy velocity by linear regression and error on the regression.

Sample Reference	Type of experiment	D_{app} (HTO) ($10^{-10} \text{ m}^2/\text{s}$)	D_{app} (^{125}I) ($10^{-10} \text{ m}^2/\text{s}$)	η (HTO)	η (^{125}I)	η_{tot}	Anisotropy on ($D_{\text{app}}^{\text{HTO}}/D_{\text{app}}^{\text{I}^{125}}$)
Boer-Pa2	TD1**	2.15 ± 0.07	2.87 ± 0.07	1.11 ± 0.09	0.11 ± 0.01		
	TD2**	5.05 ± 0.08	2.46 ± 0.07	0.27 ± 0.01	0.21 ± 0.01		
	PI	4.32 ± 0.89	-0.79 ± 0.75	$0.34 \pm 0.01^*$	$0.21 \pm 0.02^{***}$		
	BtB**	5.33 ± 0.71	3.06 ± 0.59	0.28 ± 0.01			
	Average	5.18 ± 0.17	2.80 ± 0.31	0.30 ± 0.05		$0.28 \pm 0.01^{**}$	
Boer-Pes	TD1*	3.54 ± 0.40	1.90 ± 0.08	0.36 ± 0.03	0.13 ± 0.03		
	TD2*	3.43 ± 0.19	1.52 ± 0.15	0.28 ± 0.08	0.30 ± 0.01		
	PI	$3.46 \pm 0.21^*$	$1.37 \pm 0.09^{***}$	$0.33 \pm 0.01^*$	$0.17 \pm 0.01^{***}$		
	BtB*	3.30 ± 0.40	1.25 ± 0.15				
	Average	3.43 ± 0.10	1.51 ± 0.28	0.32 ± 0.04		$0.30 \pm 0.01^*$	1.5/1.9
Putte-Pas	TD1*	5.53 ± 0.01	3.43 ± 0.06	0.33 ± 0.01	0.090 ± 0.001		
	TD2*	5.12 ± 0.39	2.68 ± 0.03	0.31 ± 0.02	0.17 ± 0.01		
	PI	$5.76 \pm 0.22^*$	$2.66 \pm 0.17^{***}$	$0.40 \pm 0.01^*$	$0.17 \pm 0.01^{***}$		
	BtB*	6.95 ± 0.05	2.09 ± 0.59				
	Average	5.84 ± 0.79	2.80 ± 0.57	0.35 ± 0.04		$0.37 \pm 0.01^*$	
Putte-Pes	TD1*	2.54 ± 0.16	1.44 ± 0.06	0.35 ± 0.03	0.08 ± 0.02		
	TD2*	2.54 ± 0.07	0.91 ± 0.07	0.32 ± 0.02	0.28 ± 0.04		
	PI	$2.47 \pm 0.32^*$	$0.91 \pm 0.06^{***}$	$0.39 \pm 0.01^*$	$0.18 \pm 0.01^{***}$		
	BtB*	3.45 ± 0.8	1.29 ± 0.39				
	Average	2.75 ± 0.47	1.16 ± 0.25	0.35 ± 0.03		$0.34 \pm 0.01^*$	2.1/2.4

(Figures ES14 and ES15) as examples. Overall, the experimental data were consistent with the respective hydraulic conductivity of the Boom Clay plugs and with the imposed hydraulic gradient, i.e., faster breakthrough was obtained for the samples with the highest conductivity and/or for the experiment performed at the highest hydraulic gradients. The model was able, in all cases, to accurately describe the experimental results. All of the experimental and breakthrough curves as well as a case by case description of the results can be found in Durce et al., (2022b).

Fig. 2 reports the accessible porosity calculated for HTO and ^{125}I , respectively, using equation (7) as a function of the calculated Darcy velocity for all the performed experiments. The Darcy velocity is proportional to the hydraulic conductivity of the sample and the range of velocities covered varies from plug to plug, samples with an orientation

parallel to the bedding and/or silty-rich covering higher Darcy velocities. To be more representative of the variation of hydraulic conductivity, it was chosen here to plot all experiments independently and not to average the results of the plugs of the same origin and orientation. Only four pulse injection experiments were performed on the plug Boer-Pa2 instead of six as for the other plugs. The Boer-Pa2 plug showed damages after the through-diffusion experiment as revealed by an increase in its hydraulic conductivity. Though some time was allowed for self-sealing and recovery of the hydraulic conductivity, it appeared, as discussed later on, that results obtained at 10 and 12 bars were inconsistent possibly indicating a re-opening of some fractures. The pulse injections at the highest hydraulic gradients, i.e., 15 and 19 bar (corresponding to the highest apparent velocities) were consequently not

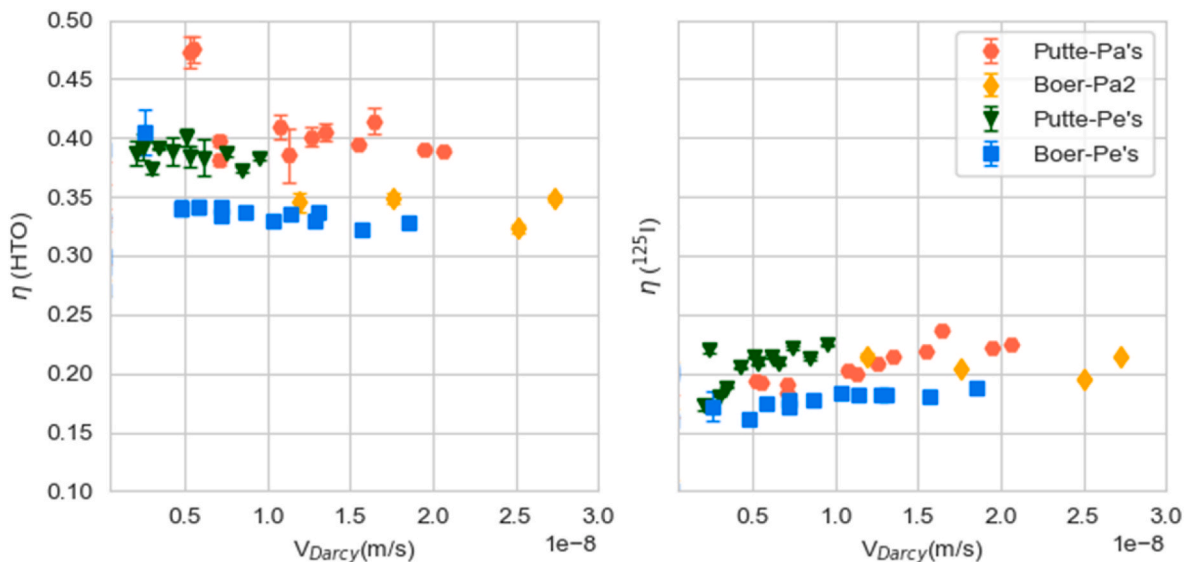


Fig. 2. Accessible porosity for HTO (left) and ^{125}I (right) as a function of Darcy velocity in pulse injection experiments calculated according to Equation (7). Uncertainties resulting from the propagation of numerical uncertainties associated with V_{app} are reported.

performed and the sample was left to reseal instead.

The values of the accessible porosity determined for HTO and ^{125}I are relatively independent of the Darcy velocity and this for all samples as also observed by Aertsens M. et al. (1999). The values are in good agreement with values previously determined from pulse injection experiments (Aertsens et al., 2004). Yet, a close look at the HTO data here reveals a slight decrease of the accessible porosity with an increase of Darcy velocity. The opposite trend is observed for ^{125}I with an increase of the accessible porosity at higher velocities. This evolution is more marked than for HTO with significant regression coefficients (Table ES7 and ES8 Electronic Supplement) and present in all the samples but Boer-Pa2. The apparent modification of the accessible porosity at higher hydraulic gradients is consistent with the decrease and increase in the accessible porosity of HTO and ^{125}I , respectively, determined by through-diffusion before and after the series of pulse injection experiments (Table 2). This suggests that an increase of hydraulic gradient could slightly affect the porosity of the Boom Clay sample. For HTO, the effect remains small enough over the investigated Darcy velocities so that the accessible porosity values can be averaged. For ^{125}I on the contrary, it was chosen, to further compare with the pure diffusion experiments, to extract the accessible porosity to zero velocity by linear regression. This was done for all samples but Boer-Pa2 for which the accessible porosity could be averaged. The results are reported in Table 4 and plotted in Figs. 6 and 7.

In general, the linear relation in Equation (1) is used to relate the dispersion coefficients to a diffusion coefficient. However, Fig. 3 and Tables ES5 and ES6 demonstrate that the corresponding linear regressions are poor. For HTO, in samples Putte-Pas, Putte-Pes and Boer-Pes, the dispersion coefficient is approximately independent from V_{app} (Fig. 3), which might justify to consider the dispersion coefficient as a constant and equal to the diffusion coefficient.

The dispersion coefficients for Boer-Pa2 increase significantly at 12 and 15 bars showing a deviating behaviour from the other plugs. A similar deviation is observed in the case of ^{125}I . Because of this deviation, which could be due to the sample fragility, the dispersion coefficients obtained on Boer-Pa2 are excluded from the discussion.

For iodide, the linear regression is better than for HTO (Table ES6) and for Putte-Pes, Putte-Pas and Boer-Pes samples, the diffusion coefficient can be extrapolated at zero velocity. Clearly the negative apparent diffusion coefficient obtained for Boer-Pa2 does not make sense, which confirms the erroneous nature of the data. The values of the dispersion length α_L determined as the slope of the linear regression and mentioned

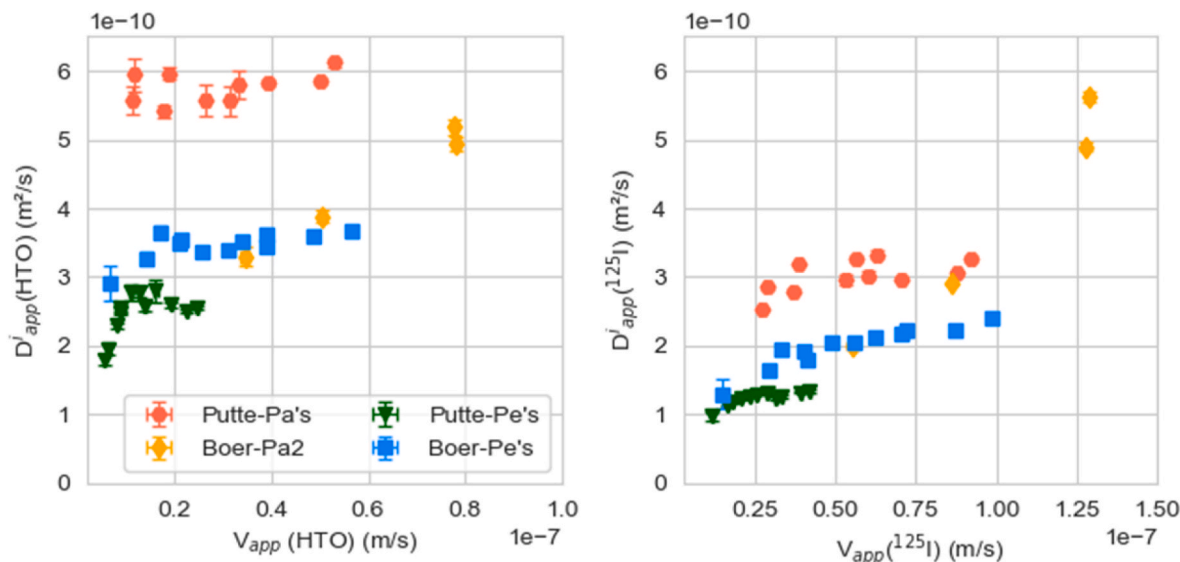


Fig. 3. Apparent dispersion coefficients for HTO (left) and ^{125}I (right) as a function of Darcy velocity in pulse injection experiments. Uncertainties resulting from the propagation of numerical uncertainties associated with V_{app} are reported.

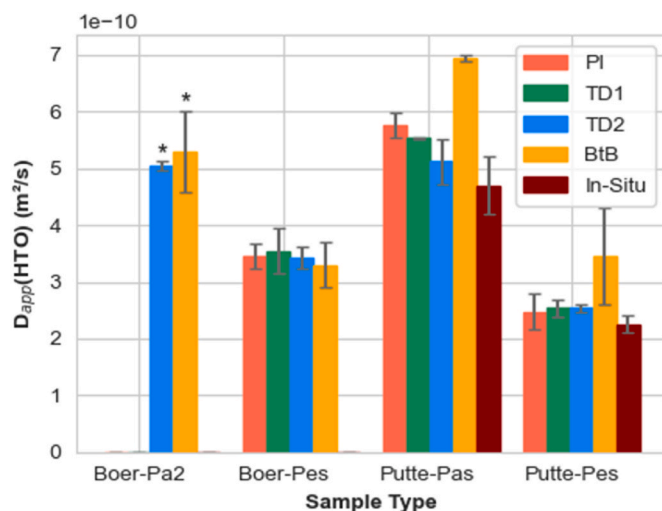


Fig. 4. Apparent diffusion coefficient of HTO determined with different experimental approaches: PI = pulse injection series, TD1 = through-diffusion experiments performed before the pulse injections, TD2 = through-diffusion experiments performed after the pulse injections, BtB = Back-to-back experiments, In-situ = average of values determined with in-situ Tribicard-3D/CP1 experiment (Aertsens M. et al., 2013). For BtB, TD1, TD2 values correspond to the average of the two plugs with the spread. * = values from one plug given with the numerical error. For PI, the values correspond to the average of the two plugs over the different hydraulic gradients and are given with the standard deviation.

in Table ES6 are similar to the values of $1.5 \cdot 10^{-3}$ m (HTO) and $4 \cdot 10^{-3}$ m (iodide) mentioned in Aertsens et al. (1999).

The poor correspondence between Equation (1) and the experimental results is not unexpected. Latrille et al. (2007) measured a HTO dispersion coefficient lower than the HTO diffusion coefficient. Although at different ionic strengths, pulse injection experiments for $\text{H}^{14}\text{CO}_3^-$ led to an impossible negative dispersion length. Besides, the dispersion length α_L is not a physical constant but depends on the scale of the measurement. Reported values for α_L vary over ten orders of magnitude (Hunt and Sahimi, 2017; Hunt et al., 2013), indicating a problem in the advection-dispersion equation (6) and the linear dispersion equation (1) (Hunt et al., 2011, 2013; Hunt and Sahimi,

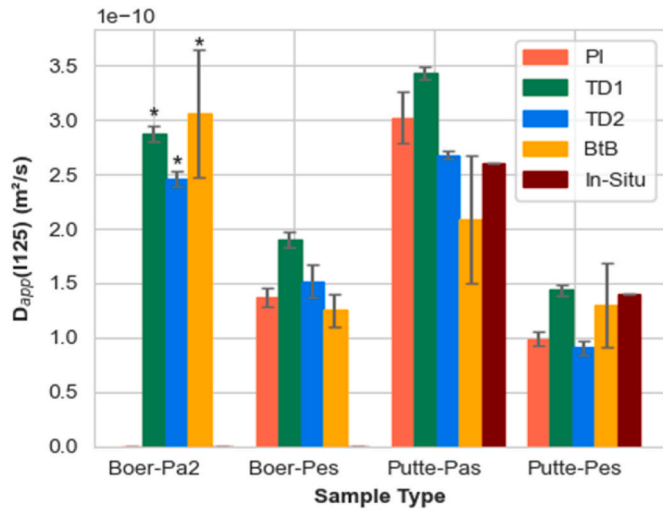


Fig. 5. Apparent diffusion coefficient of ^{125}I determined with different experimental approaches: PI = pulse injection series, TD1 = through-diffusion experiments performed before the pulse injections, TD2 = through-diffusion experiments performed after the pulse injections, BtB = Back-to-back experiments, In-situ = average of values determined with in-situ Tribicard-3D/CP1 experiment (Aertsens M. et al., 2013). For BtB, TD1, TD2 values correspond to the average of the two plugs with the spread. * = values from one plug given with the numerical error. For PI, the values correspond to the dispersion coefficient extrapolated to zero Darcy velocity by linear regression and are given with the error on the regression.

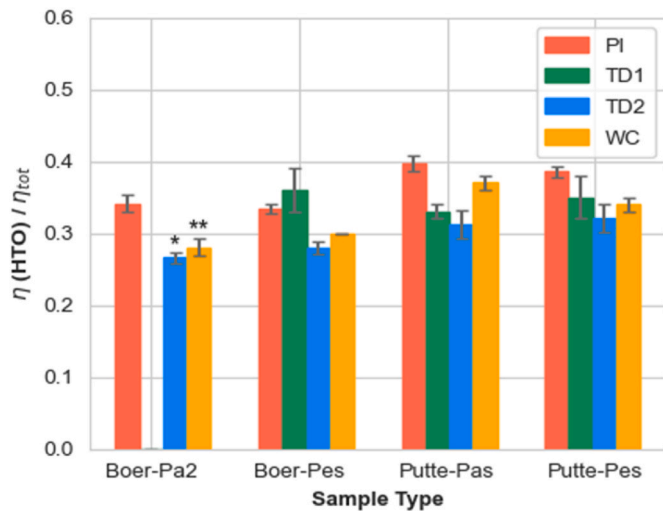


Fig. 6. Accessible porosity for HTO determined with different experimental approaches: PI = pulse injection series, TD1 = through-diffusion experiments performed before the pulse injections, TD2 = through-diffusion experiments performed after the pulse injections, WC = total porosity calculated from the water content measured at the end of the back-to-back experiments. For WC, TD1, TD2 values correspond to the average of the two plugs with the spread. * = value from one plug given with the numerical error. ** = value from one plug given with measurement error. For PI, the values correspond to the average over the different hydraulic gradients and are given with the standard deviation.

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3.3. Consistency of lab-scale transport experiments and effect of sample mineralogy and orientation

The transport parameters of HTO and ^{125}I determined in the different

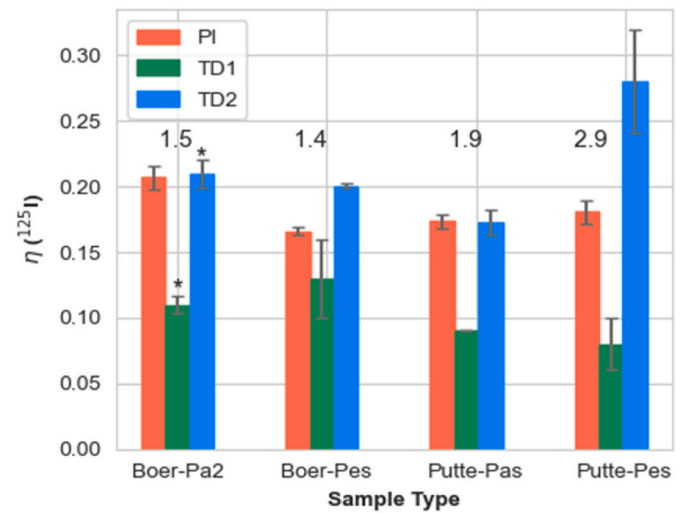


Fig. 7. Accessible porosity for ^{125}I determined with different experimental approaches: PI = pulse injection series, TD1 = through-diffusion experiments performed before the pulse injections, TD2 = through-diffusion experiments performed after the pulse injections. For TD1 and TD2, values correspond to the average of the two plugs with the spread. * = values from one plug given with the numerical error. For PI, the values correspond to the porosity extrapolated to zero Darcy velocity by linear regression and are given with the error on the regression. The indicated numbers correspond to: $(\eta(\text{PI}) + \eta(\text{TD2})) / (2 * \eta(\text{TD1}))$.

types of experiments for the different samples/orientations are summarized in Table 4. To facilitate the comparison between experiments, they are also plotted in Figs. 4–7.

3.3.1. Apparent diffusion coefficients

Figs. 4 and 5 report the apparent diffusion coefficients for HTO and ^{125}I determined by each experimental approach for the different rock samples/orientations. The D_{app} values determined here with the lab-scale experiments are compared with data obtained by in-situ experiments, which are available for the Putte Member only. Lab experiments are performed at a higher temperature (21 °C) than the in-situ experiments (16 °C in the underground). The temperature effect of the apparent diffusion coefficient is described by Arrhenius law: $D_{app} = A \exp(-E_a/RT)$ with A a constant, E_a the activation energy, $R = 8.134 \text{ J/(K mol)}$ the gas constant and T temperature (K). Taking $E_a \approx 20 \text{ kJ}$ (Van Loon et al., 2005b) leads to a 15 % increase of the apparent diffusion coefficient with respect to the in-situ value at 16 °C. Values for Boer-Pa2 determined in TD1 and in pulse injection experiments are not reported here as they are considered as erroneous.

In the case of HTO, the D_{app} values determined by pulse injection and the two through-diffusion experiments are consistent and show a variation of maximum a factor 1.1. The values are also in good agreement with the values determined in-situ and reported by Aertsens M. et al. (2013). The HTO apparent diffusion coefficients determined with the back-to-back experiments are in relatively good agreement with the other experiments for the Boer samples but are higher for the Putte samples. The reason for this deviation is not clear but could be due to the difficulties encountered in the modelling of the experimental profiles of the back-to-back experiments (activity peak not captured and difficulty to position source point).

In case of ^{125}I , the D_{app} values determined with the pulse injection experiments and the second series of through-diffusion experiments are consistent while the values from the first series of through-diffusion are systematically higher. This is indicative of a modification of the Boom Clay sample porous network occurring under the imposed hydraulic gradients. Due to electrostatic effects, anions are most sensitive to small modifications in the porosity than HTO, which could explain the effect of pressure gradient observed for iodide and not HTO. In the case of

anionic tracers, percolation experiments should therefore be performed with care as their results could not be representative of the undisturbed clay. That said, the bias remains limited and D_{app} values determined by pulse injection diverge at maximum of a factor 1.5 from the first series of through-diffusion experiments. Back-to-back experiments also show some differences with the other methods, but no systematic trend is identified and despite the experimental and modelling uncertainty associated with these experiments, the determined D_{app} values remain in the same order as the other methods. As for HTO, overall the results obtained for ^{125}I from the different lab-scale experiments are consistent with D_{app} values that do not vary for more than a factor 1.7 for each sample/orientation type. The average values for the Putte samples, as reported in Table 4, are also in good agreement with the values reported from in-situ experiments (Fig. 6).

Hence, for both tracers, the three experimental methods provide relatively consistent values of apparent diffusion coefficient. Some divergence is observed between the methods due to the experimental and modelling uncertainty for the back-to-back experiments or to a slight modification of the samples with hydraulic gradients for the pulse injection experiments. Yet with D_{app} values not varying for more than a factor 1.7 for both tracers and in good agreement, at least for the clay-rich samples, with *in-situ* experiments, it can be concluded that all the three methods are robust to estimate the diffusion coefficient of non-retarded tracers.

With an average anisotropy, defined as the ratio of $D_{app}(\text{Parallel})/D_{app}(\text{Perpendicular})$ of 2.1 for HTO and 2.4 for ^{125}I in the Putte samples and of 1.5 (HTO) and 1.9 (^{125}I) in the Boer samples (Table 4), it appears that the silty samples display a lower anisotropy than the clayey samples. Anisotropy is mostly the result of the selective orientation of the clay sheets perpendicular to the compaction axis (Keller et al., 2011; Van Loon et al., 2004) and samples with lower clay content are therefore expected to show lower anisotropy. The effect of mineralogy on the anisotropy of the Boom Clay was never explicitly investigated though Boom Clay samples from different origins were used over the years. Anisotropy of diffusion coefficients of 2–3 were reported by Aertsens et al. (2009) and Bruggeman et al. (2017) while values of 1.4–1.8 were reported by Jacops et al. (2017). The values obtained here cover therefore the range of values previously observed.

If the mineralogy seems to have an influence on the anisotropy, the effect on the diffusion coefficient is less clear. For a same orientation, the diffusion coefficients slightly differ between Boer and Putte samples, but no consistent trend emerges even though the hydraulic conductivity of the Boer samples is higher than of the Putte samples (Table 1). The same observation was made previously by Jacops et al. (2017) and Aertsens et al. (2004).

3.3.2. Accessible porosity

Figs. 6 and 7 report the accessible porosity for HTO and ^{125}I determined by each experimental approach for the different rock samples/orientations and the measured water content. The values for Boer-Pa2 for HTO determined in TD1 is not reported here as it is considered as erroneous.

Some variations are visible in the HTO accessible porosity determined by the pulse injection and the two series of through-diffusion experiments with systematically lower porosities obtained in the second series of through-diffusion experiments. This could be the result of a partial compression of the clay sheets under the hydraulic gradients imposed in the pulse injection experiment series. However, the average accessible porosity determined in pulse injection experiments is mostly higher than the first through-diffusion series and so do not show any sign of porosity reduction due to compression. The lower porosity determined in the second series of TD experiments is therefore assumed to be more the result of experimental and modelling uncertainty than to a physical modification of the Boom Clay samples. Over the three experimental sets, PI, TD1 and TD2 the accessible porosity for HTO does not vary more than a factor 1.3 for each sample/orientation. The porosity

averaged over the two through-diffusion and the pulse injection experiments (Table 4) is in good agreement with the total porosity determined from the water content of the clay cores. The total porosity of the Boom Clay samples is therefore accessible to HTO, independent of the mineralogy and the flow orientation, and both pulse injection and through-diffusion experiments are reliable methods to estimate it.

In the case of ^{125}I , a clear difference is noticed in the accessible porosity determined in the PI/TD2 series and the TD1 series (Fig. 7). Except for the Putte-Pes samples, the accessible porosity determined in the TD2 series and in the pulse injection series give consistent values. On the other hand, porosity accessible to ^{125}I is systematically lower in the TD1 series. Aertsens et al. (2020) showed that using Equation (7) to calculate the porosity accessible to anionic tracer is erroneous. It does not account for the reduction of the accessible Darcy flow and leads to an overestimation of the accessible porosity. The authors attributed the difference already previously observed in the ^{125}I accessible porosity determined by pulse injection and through diffusion experiments to this calculation/modelling error. As discussed in the previous sections, the porosity of the TD2 series is more uncertain due to the low injected activity. Nevertheless the good agreement (except for Putte-Pes) with the porosity determined in the pulse injection series suggests that the error made by using Equation (7) is not the main cause of the different ^{125}I accessible porosity obtained in PI and TD experiments. Consistently with the variation of D_{app} values, it indicates more a physical modification of the sample with hydraulic gradient. The effect of this phenomenon is more pronounced in the clay-rich Putte samples, which display a higher ratio $(\eta_{(PI)} + \eta_{(TD2)})/(2 \cdot \eta_{(TD1)})$ than Boer samples as indicated in Fig. 7. Hence if the pulse injection method remains a reliable method to determine the transport parameters for HTO (D_{app} and η), in clay rocks, it is less appropriate for anionic tracers. On the one hand, due to the lower diffusion coefficient, transport of the anionic species is more affected by the dispersion term and a proper extrapolation of the diffusion coefficient might require performing a series of experiments at different hydraulic gradients. On the other hand, anionic tracers are sensitive to a possible modification of the porosity of the sample with hydraulic gradient, which could affect the extent of the phenomenon of anion exclusion.

In line with theory, the porosity accessible to the two tracers is independent on the flow orientation and this in both the clayey and silty samples. This confirms the isotropy of the accessible porosity in clay rocks already reported by several authors (Moors, 2005; Put et al., 1990; Suzuki et al., 2004; Van Loon et al., 2004). The effect of mineralogy on the total porosity is also rather limited though a slightly higher HTO accessible porosity is fitted in the Putte sample, in agreement with the measured total porosity (Table 4). Based on the results of the TD1 series, which are believed to be the most representative values for η_{125} , the mineralogy has more influence on the iodide accessible porosity with lower values found in the Putte samples than in the Boer samples. The anion exclusion phenomenon being ‘controlled’ by the negative charge of the clay surfaces, the porosity accessible to ^{125}I is indeed expected lower in samples richer in clay, i.e., the Putte samples.

4. Conclusion

The transport parameters of solutes in Boom Clay were estimated over the years using various types of laboratory experiments. To gain confidence in the determined values, we assessed the consistency of the transport parameters obtained for the non-reactive tracers, HTO and ^{125}I , using three different types of laboratory migration experiments: through-diffusion, back-to-back and pulse injection experiments. The consistency was checked for Boom Clay samples originated from different horizons, i.e., clay-rich (Putte member) vs. silt-rich (Boeretang member), with orientation parallel or perpendicular to the bedding. The main aspect of this work, in comparison to previously published data, was the use of a single set of samples that were each subjected to the three lab-scale experiments. This approach allowed assessing

independently the role of the sample heterogeneity on the transport of ^{125}I and HTO and the role of the experimental set-up.

In the case of HTO, the three types of experiments led to a consistent set of transport parameters for each type of Boom Clay sample. Both the apparent dispersion coefficient and the accessible porosity were found independent of the pressure gradient and a single pulse injection experiment can be performed to estimate the HTO apparent diffusion coefficient. Over the three types of experiment, the average apparent diffusion coefficients and accessible porosity vary at maximum a factor 1.4 and 1.3, respectively for a same sample type. The porosity accessible to HTO was also found in good agreement with the total porosity calculated from the water content. This showed that the total porosity of Boom Clay is accessible to HTO and that both pulse injection and through-diffusion experiments are robust methods to determine it.

The case of ^{125}I revealed the limitation of pulse injection experiments for anionic tracers. In pulse injection experiments, the dispersion coefficient of ^{125}I was influenced by dispersion and could not directly be assimilated to diffusion coefficients. In addition, both ^{125}I apparent diffusion coefficient and accessible porosity were found affected by the sequence of imposed hydraulic gradients. This revealed a physical modification of the Boom Clay sample occurring during the pulse injection experiments. Such an observation questions the validity of the ^{125}I transport parameters and more generally of the anionic species transport parameters determined by pulse injection or percolation experiments. It complements the conclusion of Aertsens et al. (2020) that pulse injection experiments are not well suited for anionic tracers, especially for the determination of the accessible porosity. Nevertheless, despite these shortcomings, the apparent diffusion coefficients determined for ^{125}I with the three types of experiment are globally consistent and vary only within a factor 1.7 for a same sample type.

Transport anisotropy was consistently observed in all the types of experiment and for both HTO and ^{125}I . The anisotropy was found sensitive to the mineralogy of the Boom Clay samples, with a higher anisotropy observed in the samples richer in clay. On the other hand, mineralogy was not found to significantly affect the apparent diffusion coefficients of the tracers under the same orientation. Slightly higher total porosity and lower ^{125}I accessible porosity were noticed in the clay-rich samples highlighting the role of the clay matrix on the porous network of Boom Clay.

The present study allowed a clear comparison of three types of laboratory experiments commonly used in the context of radioactive waste for the determination of the transport properties of radionuclides in the host rocks. It showed that pure diffusion experiments (through-diffusion, back-to-back) and pulse injection experiments lead to consistent results for neutral non-reactive tracers as HTO. A comparison with data from *in-situ* experiments also revealed that all the three types of experiment provide transport parameters representative of *in-situ* behaviour. The present work also showed that pure diffusion experiments are reliable methods for the determination of transport parameters of anionic tracers. On the other hand, due to the effect of the imposed pressure gradient, pulse injections and by extension percolation experiments should rather be avoided in the case of anionic tracers. This work did not tackle cationic radionuclides and a similar study for retarded tracers would complete the present work and give a clear picture on the consistency of the different types of experiment.

CRedit authorship contribution statement

Delphine Durce: Writing – original draft, Methodology, Data curation, Conceptualization. **Marc Aertsens:** Writing – review & editing, Formal analysis, Data curation. **Norbert Maes:** Writing – review & editing, Project administration. **Marc Van Gompel:** Investigation. **Stéphane Brassinnes:** Writing – review & editing, Project administration, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.apgeochem.2024.106101>.

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